# nff-power-meter

Shunt-based energy measurement of ESP32 firmware operations

Experiment datasheet — Revision A

Bench rig: STM32 Nucleo-F446RE + ESP32 DevKit

2026-07-13

**Abstract**

This document specifies the bench rig used to measure the electrical energy consumed by an ESP32 device while it executes an `nff` operation — principally `nff ota`, the only `nff` command that places a sustained load (WiFi RX + flash writes) on the target. The measurand is **joules per OTA**. A 1 Ω shunt in the ESP32's ground return is sampled at 200 kSps by an STM32 Nucleo-F446RE, which accumulates charge and energy *in firmware* and reports running totals on demand. The host never integrates, and therefore cannot silently drop samples and under-report energy.

Three guards decide whether a run is reportable at all, and each exists because the rig produced a confident, plausible, *wrong* number during bring-up (§10): a DMA-overrun counter (dropped conversions), a window cross-check (a lost `ZERO` once turned a 30 s run into 91 s and 26 J), and an active continuity probe (a rig wired to *nothing* reported 590 mA and a 4.94 V rail). If any fails, no joules figure is reported.

# Contents

# 1 Scope and measurand

**Primary measurand**
Marginal energy $E_{\text{marg}}$ (J) consumed by the ESP32 module over and above its idle draw, for the duration of one `nff` command.

**Secondary**
Mean current, peak current, supply voltage, total charge, run duration.

**Purpose**
(a) reason about battery life; (b) provide a regression gate (`--max-joules`) so an agent can detect a firmware change that increases power draw.

The device under test (DUT) is the *whole ESP32 devkit board*, not the ESP32 die. The measurement therefore includes the onboard LDO loss, the USB–UART bridge, and the power LED. These are constant, so they cancel in the baseline-subtracted marginal figure (§7).

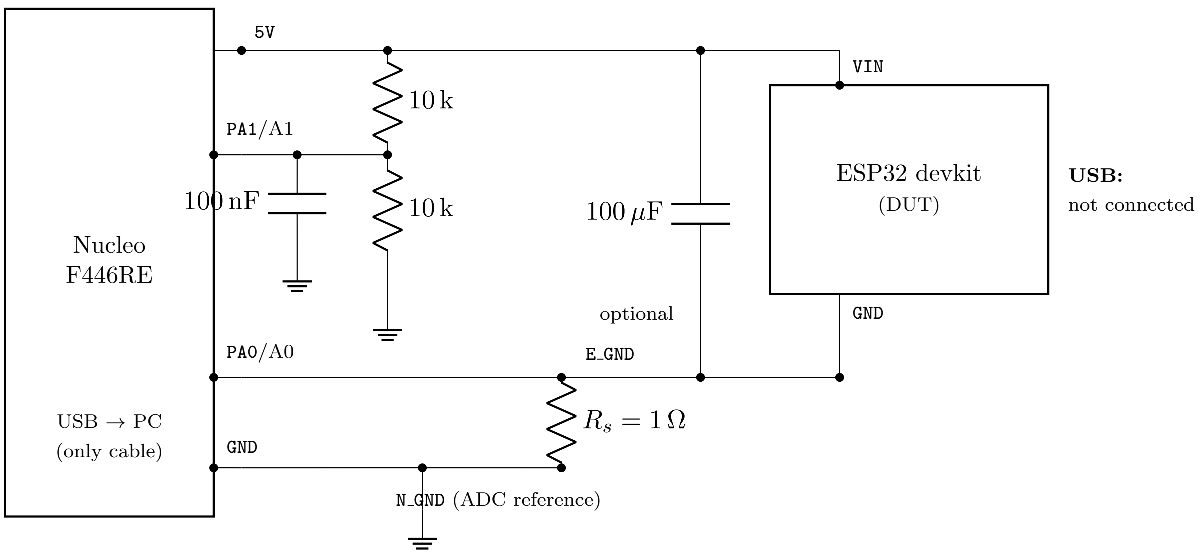
# 2 Measurement principle

The shunt is placed in the DUT's **ground return**, not its supply line. The voltage developed across it is therefore *ground-referenced* and can be read single-ended by a plain ADC input. This is the sole reason the rig needs no instrumentation amplifier.

Because the shunt is low-side, the DUT's local ground (node `E_GND`) floats above system ground by $I \cdot R_s$ — up to 300 mV at 300 mA. The voltage actually seen *across the DUT's terminals* is consequently the rail voltage minus that lift, and it is that difference — not the rail alone — that must multiply the current to obtain true power.

# 3 Schematic

## 3.1 Power path

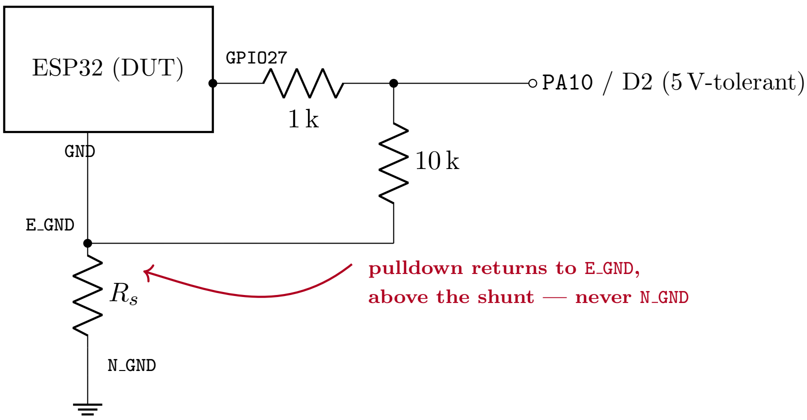



**Read it as a loop.** Current leaves the Nucleo `5V` pin, enters the DUT at `VIN`, passes through the entire board, exits at the DUT `GND` pin, and returns to `N_GND` through $R_s$. Every milliampere the DUT draws is forced through the shunt. *The DUT must have no second ground path* — see §13.

**The return-path trap.** The bulk capacitor $C_b$ returns to `E_GND` — the DUT's *own* ground, at the top of the shunt — **not** to `N_GND`. A component returned to `N_GND` carries DUT current that bypasses the shunt entirely and is therefore invisible to the meter: a silent, systematic under-read. The same rule governs the marker pulldown below. Note $C_f$ (on `PA1`) is exempt: it carries no DUT current, only ADC sample-and-hold charge, so it returns to `N_GND`.

### 3.2 Phase marker (Phase 5, optional)

Drawn separately because its ground return is the easiest thing on this rig to get wrong.




Every ESP32 GPIO goes high-impedance during reset, and an OTA *ends* in a reboot — so without the pulldown the marker line floats through precisely the phase transition being captured. The 10 kΩ holds it at a defined low; the 1 kΩ protects both pins should one end be misconfigured as an output. Returning the pulldown to `N_GND` would route ≈ 330 µA of DUT current around the shunt.

**Two markers are better than one.** A single pin distinguishes only "in a phase / not in a phase". Two pins (e.g. GPIO27 + GPIO26 into two FT inputs) give a 2-bit code — exactly the four OTA phases worth separating: download / write / verify / reboot. Same wiring cost.

## 4 Bill of materials

| Ref | Value | Notes |
|---|---|---|
| MCU | Nucleo-F446RE | Meter + host link. Only device connected to the PC. |
| DUT | ESP32 devkit (WROOM-32) | Powered exclusively from the Nucleo 5 V pin. |
| $R_s$ | 1 Ω, 0.25 W | Shunt. Ten 10 Ω in parallel is an acceptable substitute (also averages tolerance down). |
| $R_1, R_2$ | 1 to 10 kΩ | Rail divider, matched pair; 1 % preferred (ratio error is a direct gain error on energy). 10 kΩ/10 kΩ settles with ≈ 6× margin — see §6.2. Ratio is 2.0 either way. |
| $C_b$ | 100 µF | *Optional.* Bulk decoupling, **VIN→E_GND only**. |
| $C_f$ | 100 nF | *Optional.* PA1 to `N_GND`; buffers the ADC sample-and-hold. Not required at 10 kΩ, but harmless insurance. |
| $R_3$ | 1 kΩ | Marker series protection (Phase 5). |
| $R_4$ | 10 kΩ | Marker pulldown to `E_GND` (Phase 5). |
| *Reference instrument:* bench multimeter, used for calibration only. | | |

## 5 Connection table

| Nucleo | Arduino hdr | Connects to | Function |
|---|---|---|---|
| `5V` | CN6-5 | ESP32 `VIN` (= `5V`) | Supply rail |
| `GND` | CN6-6 | Bottom of $R_s$ / divider bottom | System ground, ADC reference |
| `PA0` | A0 | Top of $R_s$ = `E_GND` | Shunt sense, 200 kSps |
| `PA1` | A1 | Divider midpoint | Rail sense, $V_{\mathrm{rail}}/2$ |
| `PA10` | D2 | ESP32 GPIO27 via 1 kΩ | Phase marker (optional, 5 V-tolerant) |
| **ESP32 USB port: not connected.** See §13. | | | |

**Why PA10 and not PA4.** PA4 is a DAC pin and is *not* 5 V-tolerant. The marker's logic high arrives at up to 3.6 V (3.3 V referenced to a ground that is itself floating 0.3 V up), leaving PA4 no headroom. PA10 is an FT pin.

**Why GPIO27 on the DUT.** Not a strapping pin, not a flash pin (GPIO6–11), not input-only (GPIO34/35/36/39), no default peripheral, no boot-time glitch. GPIO13/25/26/33 are equivalent alternates. **Never GPIO12** — it is the MTDI strapping pin; held high at reset it selects a 1.8 V flash supply and the board will not boot.

## 6 Electrical design budget

### 6.1 Shunt sizing and LDO headroom

At the WiFi TX peak, $I_{\mathrm{pk}} \approx 300\,\mathrm{mA}$:

$$V_{R_s} = I_{\mathrm{pk}} R_s = 0.3\,\mathrm{V}, \qquad P_{R_s} = I_{\mathrm{pk}}^2 R_s = 90\,\mathrm{mW} \ \ (< \ 250\,\mathrm{mW}\ \text{rating}).$$

The Nucleo's 5 V pin, when the board is USB-powered, sits near 4.7 V (ST-Link protection diode drop). The DUT's AMS1117 therefore sees

$$V_{\mathrm{LDO,in}} = 4.7\,\mathrm{V} - 0.3\,\mathrm{V} = 4.4\,\mathrm{V},$$

which is at — not above — the AMS1117's worst-case dropout floor for a 3.3 V output.

**Do not increase $R_s$ beyond 1 Ω without addressing headroom.** At 2 Ω the peak drop is 0.6 V and the LDO browns out mid-OTA. If the DUT resets or resets intermittently during WiFi TX, the remedies in order of preference are: (i) drop to 0.5 Ω (two 1 Ω in parallel), accepting half the sensitivity; (ii) fit $C_b$ across VIN–E_GND; (iii) power the Nucleo from an external 7 to 12 V supply on its own `VIN`, which yields a solid 5.0 V rail.

### 6.2 ADC scaling, sampling budget, and why the divider is 1 kΩ

Both channels are read by the F446's 12 bit SAR ADC against $V_{\mathrm{DDA}} \approx 3.3\,\mathrm{V}$.

| | **PA0 (shunt)** | **PA1 (rail)** |
|---|---|---|
| Input range at ADC | 0 to 0.5 V (typ.) | ≈ 2.35 V |
| Corresponding quantity | 0 to 500 mA | ≈ 4.7 V rail |
| Full-scale before clipping | 3.3 A | 6.6 V rail |
| Raw LSB | 0.806 mV → 0.81 mA | 1.6 mV rail |
| After 256× decimation | ≈ 0.05 mA | — |
| Source impedance | ≈ $R_s$, negligible | 5 kΩ (Thévenin, 10k‖10k) |
| Sampling time | 15 cyc. | 56 cyc. |

Averaging $N = 256$ samples recovers $\log_4 N = 4$ effective bits, giving an effective 16 bit resolution — roughly 0.05 mA per count. This is what makes a bare 12 bit ADC with no front-end usable.

**Divider impedance: 10 kΩ/10 kΩ is fine**

The ADC input is an RC: the sample-and-hold capacitor ($C_{\mathrm{ADC}} \approx 4\,\mathrm{pF}$) charges through the source impedance plus the internal mux resistance ($R_{\mathrm{ADC}} \approx 6\,\mathrm{k\Omega}$). Settling to $\frac{1}{2}$ LSB at 12 bit needs $\approx \ln(2^{13}) \approx 9\tau$. For a 10 kΩ/10 kΩ divider (a 5 kΩ Thévenin source):

$$\tau = (5\,\mathrm{k\Omega} + 6\,\mathrm{k\Omega}) \cdot 4\,\mathrm{pF} \approx 44\,\mathrm{ns}, \qquad t_{\mathrm{settle}} \approx 9\tau \approx 0.4\,\mathrm{\mu s} \ll \underbrace{\frac{56}{22.5\,\mathrm{MHz}} = 2.49\,\mathrm{\mu s}}_{\text{available at 56 cyc.}}$$

— roughly **6× margin**. The 480-cycle figure sometimes quoted is the spec for a ≈ 200 kΩ source, which a resistor divider is nowhere near.

1 kΩ/1 kΩ works equally well and has more margin still, at the cost of ≈ 2.5 mA drawn from a rail that is already tight for the DUT's peaks (§6.1). That current returns to `N_GND` *without passing through the shunt*, so it does not corrupt the measurement either way. **Use whichever you have.** The ratio is 2.0 in both cases, so `KDIV` is unchanged.

**Acceptance check, not an argument.** A source impedance too high for its sampling window biases the rail reading *low*, silently — and every joule scales with the rail. So verify it: compare the rail PA1 reports (`nff power monitor`) against the multimeter. They must agree. If PA1 reads low, drop to 1 kΩ/1 kΩ or fit $C_f$.

### 6.3 Sampling budget

The ADC clock is PCLK2/4 = 22.5 MHz, and a 12 bit conversion costs ($t_{\mathrm{samp}} + 12$) ADC cycles. One scan of both ranks is therefore

$$\underbrace{(15+12)}_{\text{PA0}} + \underbrace{(56+12)}_{\text{PA1}} = 95 \text{ cycles} = 4.22\,\mathrm{\mu s},$$

against the 5.00 µs that the 200 kSps trigger allows — about 16 % headroom. If that budget is ever violated the `ovr` counter says so; it is not left to inference. *Measured on the bench:* 2 040 576 samples in 10 s ($f_s = 199\,998$ Hz), `ovr` = 0.

### 6.4 $V_{\mathrm{DDA}}$ tolerance

The Nucleo's 3.3 V regulator has a ±2 % tolerance, which appears as a ±2 % gain error on *both* channels.

**This is absorbed by the $R_{\mathrm{eff}}$ calibration (§8), not corrected independently.** Calibration solves $R_{\mathrm{eff}}$ from a measured current, so any gain error common to the chain — resistor tolerance, ADC gain, $V_{\mathrm{DDA}}$ — lands in that single constant. They are not separable, and they do not need to be.

*Not implemented:* the F446 carries a factory VREFINT constant at `0x1FFF7A2A` from which the true $V_{\mathrm{DDA}}$ could be derived independently ($V_{\mathrm{DDA}} = 3.3\,\mathrm{V} \cdot \mathrm{VREFINT\_CAL}/\mathrm{VREFINT\_DATA}$). The firmware does *not* read it, because calibration already absorbs the error. It would only earn its keep if one wanted an *uncalibrated* absolute reading, or to track $V_{\mathrm{DDA}}$ drift with temperature.

## 7 Firmware integration mathematics

Let $k$ index samples taken at $\Delta t = 5\,\mathrm{\mu s}$ ($f_s = 200$ kSps), and let $q_k$, $u_k$ be the *raw ADC counts* on PA0 and PA1. With $L = V_{\mathrm{DDA}}/4096$ the volts-per-count, $K$ the divider ratio (2.0 for a matched pair) and $R_{\mathrm{eff}}$ the *calibrated* shunt resistance (§8) — not the nominal 1 Ω:

$$I_k = \frac{q_k L}{R_{\mathrm{eff}}}, \qquad V_{\mathrm{board},k} = \underbrace{K\,u_k L}_{\text{rail}} - \underbrace{q_k L}_{\text{ground lift}}.$$

Energy then collapses into sums the firmware can keep in *exact integers*:

$$E \;=\; \sum_k V_{\text{board},k}\, I_k\, \Delta t \;=\; \frac{L^2}{R_{\text{eff}}} \cdot \frac{1}{f_s} \left( K \textstyle\sum_k u_k q_k \;-\; \sum_k q_k^2 \right), \qquad Q \;=\; \tfrac{L}{R_{\text{eff}}} \cdot \tfrac{1}{f_s} \textstyle\sum_k q_k.$$

So the STM32 accumulates only $\sum q$, $\sum q^2$, $\sum uq$, $\sum u$ and $n$ — as `uint64`, which a 10 min run at 200 kSps does not come close to overflowing — and the host multiplies by constants. A float accumulator would drift over $10^7$ samples; these do not.

**The host never integrates.** This is deliberate: a host-side integrator that misses serial frames silently under-reports energy, and the loss is invisible in the output. Here a missed frame costs nothing — the next frame carries the correct cumulative total — and a missed *conversion* cannot hide either, because it sets `ovr` (§10).

### 7.1 Baseline subtraction

A command's cost is defined as the energy *above* the cost of merely being powered on:

$$P_{\text{idle}} = \left.\frac{E(t_1) - E(t_0)}{t_1 - t_0}\right|_{\text{DUT idle, WiFi connected}}, \qquad \boxed{E_{\text{marg}} = E_{\text{total}} - P_{\text{idle}} \cdot T_{\text{cmd}}}$$

$E_{\text{marg}}$ is the reported headline figure. Because the LDO loss, USB–UART bridge and power LED are present in both terms, they cancel.

## 8 Calibration procedure (Phase 0)

**Do not skip, and repeat after *any* rewiring.** Breadboard contact resistance is 10 to 100 mΩ *per contact* and sits in series with a 1 Ω shunt. It is not negligible, and it changes when a wire is re-seated.

1. Wire the rig per the schematic. Flash the meter firmware.
2. Hang a known resistive load on the DUT's 3.3 V rail — 100 Ω gives ≈ 33 mA.
3. Read the true current $I_{\text{DMM}}$ with the multimeter *in series* with that load.
4. Read the current $I_{\text{rep}}$ the STM32 reports, assuming nominal $R_s = 1\,\Omega$.
5. Solve for the effective shunt and persist it:

   $$R_{\text{eff}} = R_{\text{nom}} \cdot \frac{I_{\text{rep}}}{I_{\text{DMM}}}$$

   This single constant absorbs the resistor's tolerance, the ADC gain error, *and* the breadboard contact resistance.
6. Repeat for the divider: measure the rail with the DMM, compare against $2\,V_{\text{PA1}}$, and store the ratio correction.

**Acceptance criterion.** After calibration, the STM32 and the multimeter must agree to within a few percent on a purely resistive load. *If they do not, stop* — nothing built on top of this rig is worth anything.

# 9 Host interface

ST-Link Virtual COM Port, 921 600 baud, 8N1, line-oriented ASCII.

| Dir | Command | Meaning |
|---|---|---|
| H→M | `PING` | Liveness / identification probe. |
| H→M | `INFO` | Firmware identity, $f_s$, calibration constants. |
| H→M | `ZERO` | Reset the accumulators and the clock. Acknowledged. |
| H→M | `SNAP` | Emit one frame *now*. |
| H→M | `STREAM <0\|1>` | Free-run frames at 10 Hz (live `monitor` view only). |
| H→M | `WIRECHECK` | Actively probe whether PA0/PA1 are connected (§10). |
| H→M | `CAL <micro_ohm>` | Set $R_{\text{eff}}$. |
| H→M | `KDIV <milli>` | Set the divider ratio ×1000 (`2000` = any matched pair). |
| H→M | `VDDA <mv>` | Set the assumed ADC reference. |
| M→H | (frame) | One JSON object per line. |

**Calibration lives on the host** (`~/.nff/config.json`) and is pushed down with `CAL`/`KDIV`/`VDDA` on every connect. The meter is stateless across resets; there is one source of truth.

**Frames are emitted on demand, not free-running.** A meter that chatters continuously leaves the host unable to say exactly when accumulation stopped relative to the profiled command exiting — it would over-attribute up to a frame's worth of idle energy to it. `measure` therefore drives `ZERO` → run → `SNAP`.


```
{"t":10200,"n":2040576,"ovr":0,"sq":141600000,"sqq":9832000000,
 "suq":4.39e11,"su":6.33e9,"qmax":410,"fs":200000,"vdda":3300,
 "r":1000000,"kdiv":2000}
```


The frame carries *raw integer sums*, not derived joules. Two reasons. It keeps the accumulation exact — a float accumulator drifts over $10^7$ samples — and it means a *re*-calibration can re-derive the energy of a past run without re-measuring it.

# 10 How the rig refuses to lie

Every guard here exists because the rig produced a confident, plausible, *wrong* number during bring-up. None is hypothetical, and `ovr` alone catches none of the last two.

## 10.1 `ovr` — dropped conversions

A missed conversion means the accumulators are an under-count. If `ovr` > 0, or the profiled child process exits non-zero, the host reports `ok: false` and does **not** present a joules figure. A plausible-but-wrong energy number is worse than no number, because it will be trusted.

## 10.2 Window cross-check — a lost `ZERO`

A `ZERO` that never lands leaves the meter accumulating *from boot*, so the frame describes the board's uptime rather than the command.

**Observed:** a 30 s run reported 91 s and 26 J — with `ovr` = 0, a perfectly sane 58 mA, and `ok: true`. The overrun counter cannot catch this: no samples were lost. The *window* was wrong.

`ZERO` is now acknowledged, and the host cross-checks (a) the meter's own elapsed clock against how long it actually waited, and (b) that the samples delivered match the claimed $f_s$ — if the ADC is not running at $f_s$, every $\Delta t$, and so every joule, is scaled wrong.

### 10.3 `WIRECHECK` — a rig wired to nothing

The nastiest failure, and it defeated two designs before this one.

> **An unconnected ADC pin does not read zero — it holds charge.** With *nothing attached at all*, the meter reported 590 mA and a rail of 4.94 V. Both are entirely plausible. Any passive range check waves them straight through.

An internal pull-up/pull-down probe fails too: it reported both pins "connected" with nothing attached, because **the F446's ADC cannot read a pad while the GPIO is in digital-input mode** (the analog switch is open), and the pulls are *disabled* in analog mode — so there is no configuration in which both work.

What survives is to **drive the node, release it to analog, and read it immediately**. A real source restores its own voltage in nanoseconds (the divider is a few-kΩ source; the shunt node is a near short to ground). A floating node keeps the charge. Measured: a floating PA0 stayed 1 746 counts off, a floating PA1 2 425 counts off; a wired node moves under 100.

For the divider, both tests must pass: it must *read* like a 5 V rail *and* spring back. Either alone can be passed by luck — the bench's floating PA1 sat at a convincing 4.94 V — but it restored to 0.5 V, and nothing genuinely connected does that.

> **PA0 is only ever driven LOW.** Driving it high would put the GPIO against the 1 Ω shunt straight to ground — ≈ 65 mA, well past the pin's 25 mA rating. PA1 is safe to drive both ways: the divider's series resistance limits the current to a few mA.

## 11 Capability envelope

| | Quantity | Remark |
|---|---|---|
| yes | Active-mode current, joules per OTA | The target measurand. |
| yes | Idle / connected-but-quiet baseline | Required for marginal subtraction. |
| yes | Peak current ($i_{\max}$) | Smoothed if $C_b$ is fitted; $Q$ and $E$ remain exact. |
| no | Deep sleep | 10 µA×1 Ω = 10 µV, roughly 100× below the breadboard noise floor. Unrecoverable without a real front-end (INA228, PPK2). |
| no | `nff flash`, `nff monitor` | Require the DUT's USB, which shorts the shunt. |
| *Caveat:* the measurement is whole-board, not ESP32-die. Constant, so it cancels in $E_{\text{marg}}$. | | |

## 12 Error budget

| Source | Raw | Residual | Treatment |
|---|---|---|---|
| Shunt tolerance (±5 %) | 5 % | < 0.5 % | Absorbed by $R_{\text{eff}}$ calibration. |
| Breadboard contact resistance | 1 to 10 % | < 1 % | Absorbed by calibration; **drifts on re-seat**. |
| $V_{\text{DDA}}$ regulator (±2 %) | 2 % | < 0.5 % | Absorbed by $R_{\text{eff}}$ calibration (*not* VREFINT — see §6.4). |
| Divider ratio | 1 % | < 0.5 % | Calibrated against the DMM. |
| Divider source impedance | — | ≈ 0 | 1 to 10 kΩ both settle with ≈ 6× margin at 56 cycles (§6.2). Verify the rail against the DMM. |
| Shunt tempco (self-heating, 90 mW) | — | ≈ 1 % | Uncorrected. Carbon film ≈300 ppm/°C. |
| ADC INL/DNL | — | ≈ 0.2 % | Uncorrected, small. |
| Aliasing of WiFi TX spikes | — | sample-rate bound | Mitigated by $f_s$ = 200 kSps; $C_b$ trades peak fidelity for stability. |
| Dropped conversions | — | **detected** | `ovr` counter → `ok: false`. |
| Wrong accumulation window | — | **detected** | `ZERO` ack + host/meter clock cross-check. |
| Rig not actually wired | — | **detected** | `WIRECHECK` drive-and-release probe. |
| **Expected overall** | | ≈ 2 to 3 % | On active-mode energy, post-calibration. |

The rig is designed for *relative* fidelity: a 2 % absolute error is irrelevant to the regression gate, which compares an OTA against a previously measured OTA on the same, un-rewired rig.

## 13 Operating rules

**R1. The ESP32 must not be plugged into the PC's USB.** A USB cable ties the DUT's ground to PC ground, which shorts the shunt out. A reading of ≈ 0 mA during a known-active phase is the signature of exactly this mistake. It is also the reason `nff flash` and `nff monitor` cannot be profiled here.

**R2. Only the Nucleo connects to the PC.**

**R3. $R_s$ = 1 Ω, 0.25 W.** Do not exceed 1 Ω without bulk capacitance — the DUT's LDO will brown out (§6.1).

**R4. Recalibrate after any rewiring.** Contact resistance is in the measurement path.

**R5. All DUT return currents pass through the shunt.** $C_b$ and the marker pulldown return to `E_GND`, never `N_GND`.

## 14 Verification sequence

Run in order. Each step gates the next.

1. **Resistive-load check.** 100 Ω on the DUT's 3.3 V rail. `nff power monitor` must agree with the multimeter to within a few percent. *If it does not, nothing else counts.*
2. **Baseline.** `nff power measure --for 30`, DUT idle and WiFi-connected. Expect tens of mA. A reading near zero means the shunt is shorted — check R1.

3. **The real number.** `nff power measure --during "nff ota ..."` → joules per OTA. Confirm `ovr` = 0.
4. **Regression gate.** Re-run with `--max-joules` set just below the measured value; confirm exit status 1.
5. **Agent loop.** Call the `power_measure` MCP tool; confirm the agent receives a structured verdict.

**Reporting.** State the measured joules-per-OTA plainly, including whether `ovr` was clean. If the rig cannot be made to agree with the multimeter, say so — do not report a number.

## 15 Measurement log

| Date | $R_{\mathbf{eff}}$ | $P_{\mathbf{idle}}$ | $E_{\mathbf{marg}}$ | $i_{\mathbf{max}}$ | ovr | Notes |
|---|---|---|---|---|---|---|
| 2026-07-14 | **1.0 Ω nom.** | 206 mW | 5.37 J | 314 mA | 0 | 1 MiB OTA, `esp32-nffload`. $3.06 \times 10^6$ samples. **Uncalibrated** — resistor measures 1.1 Ω. |
| 2026-07-14 | — | — | — | — | 0 | $f_s$ verified: 2 040 576 samples in 10 s = 199 998 Hz. |
| 2026-07-13 | — | — | — | — | — | `WIRECHECK` bring-up: wired to *nothing*, the rig read a plausible 590 mA and a 4.94 V rail. |

*Revision A.1 — rig* ***built, debugged and in service*** *(2026-07-14). It has produced real numbers: see* `nff-power-characteristics.tex`*.* ***Phase 0 calibration still not performed****, so absolute figures carry 10 to 30 %; relative figures are good to a few percent. The measurement log below is real.*